MILESTONE – 3 TECHNICAL REPORT

# AI-Based Patient Appointment System

Milestone 3: System Architecture, Technology Stack & Comprehensive Module Design


PROJECT TITLE
AI Patient Appointment System

SUBMISSION MILESTONE
Milestone 3 (Architecture & Design)

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## 1 Executive Summary & Milestone 3 Scope

Building upon the clinical datasets, exploratory analyses, and preprocessing pipelines formalized in **Milestone 2**, **Milestone 3** establishes the holistic software architecture and subsystem module design of the **AI-Based Patient Appointment System**. In modern healthcare delivery, operational inefficiencies frequently arise because patient registration, symptom intake, clinical triage, doctor availability tracking, and medical consultation histories exist in disjointed silos.

Milestone 3 synthesizes these disparate components into a cohesive, high-performance, asynchronous multi-tier architecture. It explicitly formulates the system contracts, database schemas, API pathways, and data flows connecting the intelligent conversational chatbot to patient profiles, physician appointment books, and provider consultation portals.

| 8 | 3-Tier | Dual | < 100ms |
|---|---|---|---|
| CORE MODULES DESIGNED | DECOUPLED ARCHITECTURE | AUTH LAYER (GOOGLE + OTP) | API GATEWAY TARGET |

**ARCHITECTURAL DESIGN MANDATE**

*The architectural blueprint ensures strict decoupling between clinical artificial intelligence, authentication security, and transactional appointment booking. Clinical AI recommendations are architected as assistive, non-prescriptive preliminary assessments requiring attending physician validation.*

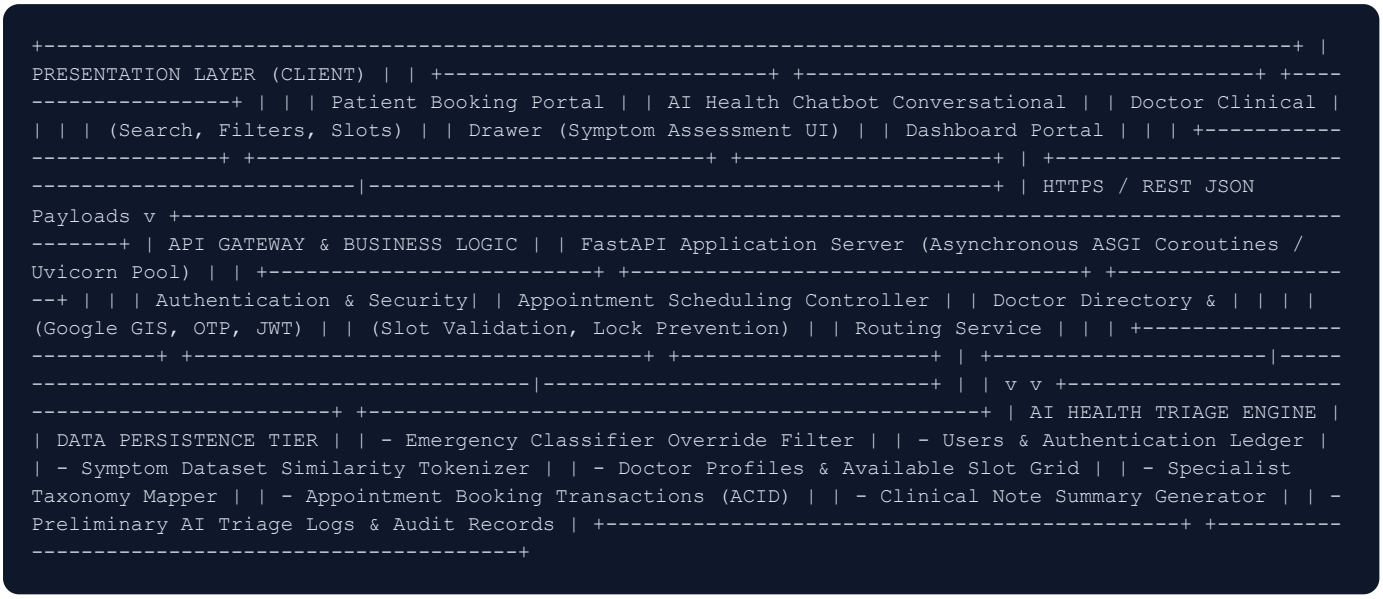
## 2 Technology Stack & Architectural Decisions

The selected technology stack reflects modern engineering principles: asynchronous non-blocking I/O, strict cryptographic security, type safety, and real-time responsiveness.

| Layer | Selected Technology | Rationale & Architectural Advantage |
|---|---|---|
| **Frontend Client** | REACT 18 + VITE / SPA | Sub-second initial paint, reactive state management for real-time conversational chat, modular component architecture, and responsive glassmorphic styling. |
| **API Gateway** | FASTAPI (PYTHON 3.10+) | High-throughput ASGI server (Uvicorn), native Python async coroutines, automatic OpenAPI documentation, and strict Pydantic v2 data serialization. |
| **AI Triage Engine** | NLP SYMPTOM MATCHER | Dual-stage processing combining deterministic emergency keyword overrides with vectorized symptom-disease similarity scoring against validated clinical datasets. |
| **Authentication** | GOOGLE GIS + JWT + BCRYPT | Stateless cryptographic security using HMAC-SHA256 JWT tokens, Bcrypt password hashing (cost factor 12), and federated Google OAuth 2.0 Identity Services. |
| **Data Persistence** | RELATIONAL SCHEMA & JSON | ACID compliance for appointment transactions to completely prevent double-booking collisions, coupled with serialized records for audit trails. |
| **Communication** | REST API + ASYNC WORKERS | Stateless JSON payload endpoints for doctor listings, symptom assessment, slot reservation, and real-time dashboard updates. |

## 3 End-to-End Multi-Tier System Architecture

The architecture follows a decoupled N-Tier pattern separating presentation, application business logic, artificial intelligence heuristics, and persistent data storage.




**PRESENTATION LAYER**

Provides intuitive user journeys for both patients and healthcare providers. Implements zero-refresh reactive updates, interactive symptom questioning, calendar slot pickers, and medical summaries.

**APPLICATION GATEWAY**

Acts as the secure orchestrator. Validates inbound payloads with Pydantic schemas, manages role-based access, coordinates AI triage requests, and executes atomic booking transactions.

## 4 Comprehensive Module Decomposition

The system is decomposed into eight highly specialized functional modules with clearly delineated boundaries, input/output contracts, and responsibilities:

### 1. Patient Registration & Dual-Auth Service

Provides seamless patient onboarding via Google OAuth 2.0 (Google Identity Services) or Mobile/Email OTP verification. Employs Bcrypt password hashing and issues signed JWT bearer tokens for secure session maintenance.

### 2. AI Health Chatbot Conversational Drawer

An interactive conversational interface allowing patients to describe complaints in unstructured natural language. Guides dialogue, prompts for duration/severity, and delivers structured preliminary recommendations.

### 3. Symptom Assessment & Emergency Engine

Evaluates symptoms against high-risk clinical triggers (e.g. cardiac distress, stroke symptoms). Non-emergency queries are matched against the medical dataset to identify probable health conditions and severity tiers.

### 4. Doctor Recommendation & Specialist Router

Translates the evaluated condition into targeted medical specialties (Cardiology, Dermatology, Neurology, Orthopedics, Pulmonology, General Medicine) to eliminate patient self-triage misdirection.

### 5. Doctor Availability & Slot Management Matrix

Maintains real-time consultation calendars for certified doctors. Organizes schedules into distinct time slots, accounts for existing bookings, and dynamically removes occupied intervals.

### 6. Appointment Scheduling & Conflict Engine

Executes atomic booking transactions. Binds patient profiles, doctor records, dates, and slots. Strictly enforces uniqueness constraints to prevent concurrent double-booking anomalies.

### 7. Patient Records & Consultation History

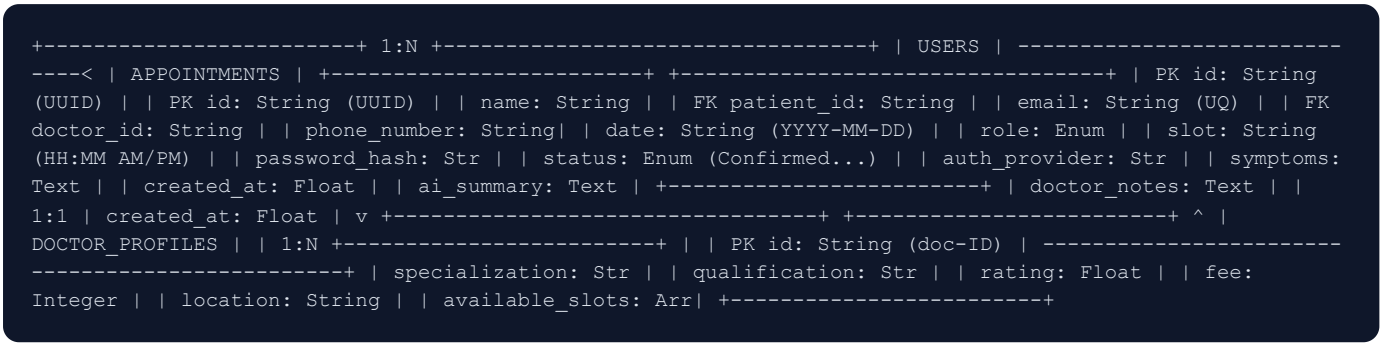
Maintains patient health records, historical appointment statuses (Confirmed, Completed, Cancelled), reported symptoms, and AI triage summaries for longitudinal continuity of care.

### 8. Doctor Dashboard & Verification Portal

A specialized portal for medical practitioners to review upcoming queues, inspect AI preliminary summaries, add official clinical consultation notes, and verify treatment plans.

## 5 Database Architecture & Relational Entity Model

The persistence model enforces third normal form (3NF) relational integrity. The design guarantees ACID compliance for appointment transactions while facilitating rapid retrieval for doctor searches and triage history.

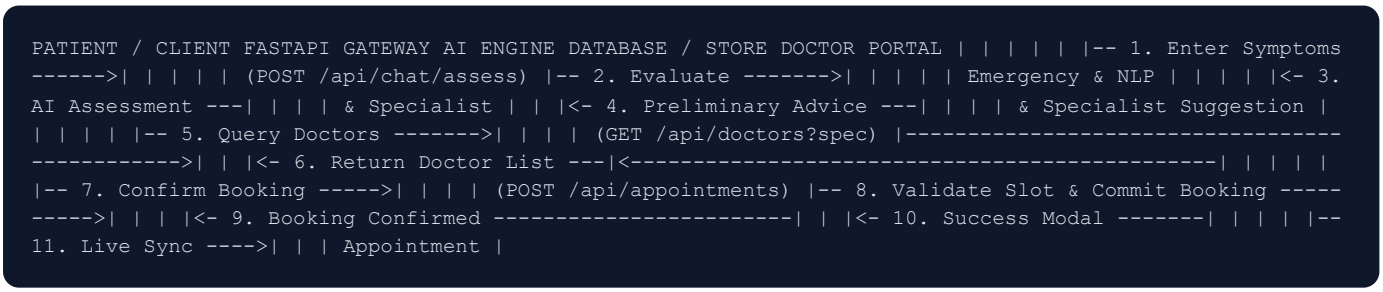



### Entity Schema Specifications

| Table Name | Primary Key | Foreign Keys | Key Attributes & Constraints |
|---|---|---|---|
| `users` | `id` (UUID) | None | `email` (Unique), `role` (patient/doctor/admin), `password_hash`, `phone_number`. |
| `doctor_profiles` | `id` (e.g. `doc-101`) | `user_id` → `users.id` | `specialization`, `qualification`, `rating` (0.0–5.0), `fee`, `available_slots`. |
| `appointments` | `id` (e.g. `apt-1001`) | `patient_id`, `doctor_id` | `date`, `slot`, `status` (Confirmed/Completed/Cancelled), `symptoms`, `ai_summary`. |
| `otp_records` | `phone_number` / `email` | None | `otp_hash`, `attempts` (Max 5), `expires_at` (5 min window), `is_consumed`. |

## 6 Communication Flows & Sequence Workflows

The interaction between modules is governed by asynchronous RESTful sequences ensuring prompt response times even under high concurrent load:




> **CRITICAL SAFETY OVERRIDE PROTOCOL**
>
> If step 2 detects acute emergency markers (e.g. severe chest pain or stroke symptoms), the sequence immediately terminates regular appointment scheduling and displays an urgent directive urging the patient to visit the nearest Emergency Room or call emergency medical services.

## 7 Clinical Safety, Security & Ethical Boundaries

### 🛡️ Medical Non-Replacement Directive

The AI engine is strictly designed as a clinical decision-support and triage tool. Every chatbot interaction explicitly incorporates standard disclaimers stating that AI guidance is preliminary and cannot replace the examination, diagnosis, or prescription of a qualified healthcare professional.

### 🔐 Role-Based Access Control (RBAC)

Stateless JWT tokens encode cryptographically signed user roles ( `patient` , `doctor` , `admin` ). API route dependencies verify permissions, preventing unauthorized access across patient health records and doctor schedule configurations.

### 🔒 Data Privacy & Confidentiality

Passwords are encrypted using Bcrypt hashing with salted cost factors. In-transit data is secured via TLS/HTTPS, and patient symptom histories remain segregated within secure backend boundaries.

### ⚡ High-Availability & Fallback Design

If the AI similarity matcher encounters unstructured or highly ambiguous inputs below the confidence threshold (0.15), it safely routes the patient to a General Physician rather than guessing an unverified specialty.

## 8 Milestone 3 Deliverables & Readiness for Milestone 4

COMPLETED IN MILESTONE 3

- **Architecture:** End-to-end multi-tier system blueprint and decoupled topology.
- **Tech Stack:** Finalized React 18, FastAPI, Uvicorn, and Python NLP engine.
- **Module Contracts:** Defined interfaces for 8 functional subsystems.
- **Data Model:** Normalized Entity-Relationship (ER) relational database schemas.
- **Safety Protocols:** Integrated clinical emergency filters & statutory disclaimers.

ROADMAP FOR MILESTONE 4

- **Conversational Chatbot:** Full development of reactive chat drawer with NLP engine.
- **Recommendation Engine:** Dynamic doctor matching based on patient symptoms.
- **Appointment Engine:** Slot booking calendar and real-time conflict locking.
- **Doctor Verification:** Doctor-in-the-loop consultation review workflow.

### 📌 Project Repository & Reference

**GitHub Repository:** https://github.com/LokendraDev06/AI-Based-Patient-Appoinment-System